

Advancing the EZIE Observing System Simulation Experiment (OSSE) for Auroral Space Weather Analysis

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Abstract—This work advances the EZIE Observing System Simulation Experiment (OSSE), developed to study auroral electrojets and their associated magnetic perturbations. We implemented a complete end-to-end data processing pipeline in Python, beginning with Level-0 quality screening to filter out calibration and reorientation intervals, ensuring reliable satellite observations. At Level-1, we constructed optimized merged atmospheric files by combining IGRF, NRLMSIS, and MERRA-2 model datasets, incorporating both realistic neutral atmospheric variables and colocated background magnetic fields. Improvements in vectorization, parallelization, and lazy-loading reduced computational costs while maintaining physical accuracy. We further developed and applied a robust vertical polarization axis alignment routine to reconcile differences between simulated and observed Stokes parameters. At Level-2, we developed a custom Gauss–Newton retrieval framework to jointly estimate magnetic field and temperature profiles from EZIE radiance observations. Initial retrieval experiments successfully identified magnetic field perturbations that reproduced independent ground-based magnetometer data, demonstrating the accuracy and robustness of the pipeline. Together, these advances provide a scalable and efficient framework for analyzing EZIE satellite observations and extracting geophysically meaningful information on auroral current systems.

1. Background

The Electrojet Zeeman Imaging Explorer (EZIE) is a heliophysics mission focusing on how the Sun influences the space weather near Earth. EZIE consists of 3 SmallSat satellites designed to sample radiances that are affected by auroral electrojets. These electrojets are electrical currents in the Earth's Magnetosphere around 60 to 90 miles (96.56064 to 144.84096 km) above Earth's magnetic poles. The EZIE mission launched on March 15 2025.

To understand what the Electrojet Zeeman Imaging Explorer does, we first have to start with the Sun, Magnetosphere and Earth's Upper Atmosphere:

1.1. Sun, Magnetosphere & Earth's Upper Atmosphere

The Sun emits ≈ 1.9 million metric tons of charged particles in every direction which we call the solar winds. These winds are moving at speeds of ≈ 1 million mph and are capable of erode an atmosphere, as seen with planets like Venus and Mars.

However, the Earth has a magnetic field and the particles flow around it to form tear-drop regions, which are known as the Earth's magnetosphere. This part of the

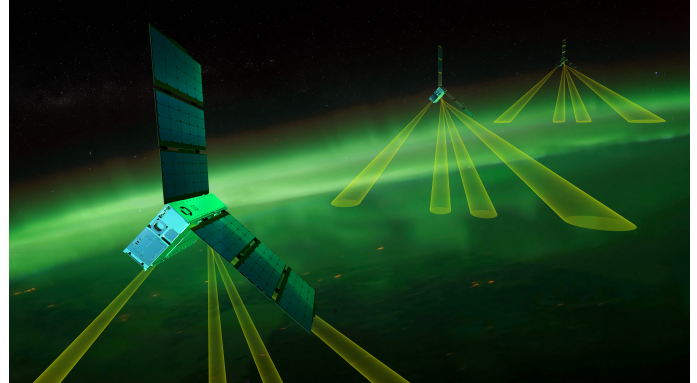


Figure 1. Depiction of NASA's EZIE (Electrojet Zeeman Imaging Explorer) satellites sampling the atmosphere in the presence of auroral electrojets. The CubeSats observe the aurora by detecting polarized electromagnetic emissions, helping scientists to better understand space weather and its interaction with Earth's magnetic field.(Image credit: EZIE APL website [3]).

atmosphere is primarily where the Earth's magnetic field exists.

The solar winds end up occasionally pushing energy into the Earth's magnetosphere, which ends up pulling it away from the Earth to form the right stretch as shown in Figure 2. Eventually, as the energy from these winds accumulates, the magnetosphere releases this energy in an event known as magnetic reconnection. This energy and particles that are released then traverse down the magnetic field lines towards the Earth's upper atmosphere.

In the upper atmosphere, as the energy and particles flow into it, the upper sphere absorbs the energy and creates the space weather phenomena commonly called the Northern lights. These effects can cause issues like damaging space crafts, causing blackouts and even disrupting our everyday life.

The flow of solar energy and particles creates an electric circuit that flows along the magnetic field lines and a small but critical part of this circuitry is known as the auroral electrojets. which EZIE aims to better understand. We will cover more of that in the next section.

1.2. Electrojets

The Earth's auroral electrojets are the electrical currents that travel through the planet's magnetic poles. These currents exist at around 65 miles (100 km) above the surface. They push roughly 1 million amps of electrical

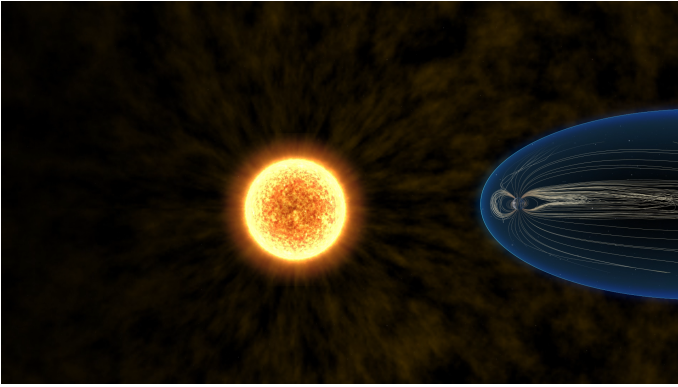


Figure 2. Illustration of solar wind interactions with Earth's magnetosphere, illustrating how charged particles from the Sun are deflected by Earth's magnetic field (Image credit: EZIE APL website [3]).

charge around the poles every second. Due to the nature of following currents, this induces a large magnetic perturbation on the ground. The goal of EZIE is to understand these disturbances and advance our understanding of their structure.

To measure the magnetic field vector perturbations induced by these electrojets, EZIE makes use of the Zeeman effect.

1.3. Zeeman Effect

The radiometers on each EZIE satellite measure the oxygen (O_2) spectral absorption line at 118.75 GHz. Molecular oxygen has a unique characteristic: in the presence of a magnetic field the absorption line at 118.75 GHz splits into multiple components. The degree and the pattern of this spectral line splitting depends on the strength and orientation of the magnetic field in the atmosphere.

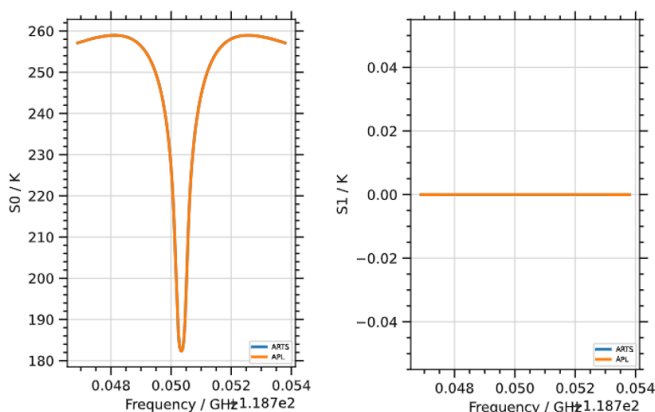


Figure 3. Simulated spectra of the first two Stokes vector elements without the presence of a magnetic field. S_0 (left) shows the total intensity, while S_1 (right) represents the difference between vertical and horizontal polarization.

In this project we are trying to derive the small magnetic field perturbations in the Earth's magnetic field and the Auroral Electrojet using these spectral

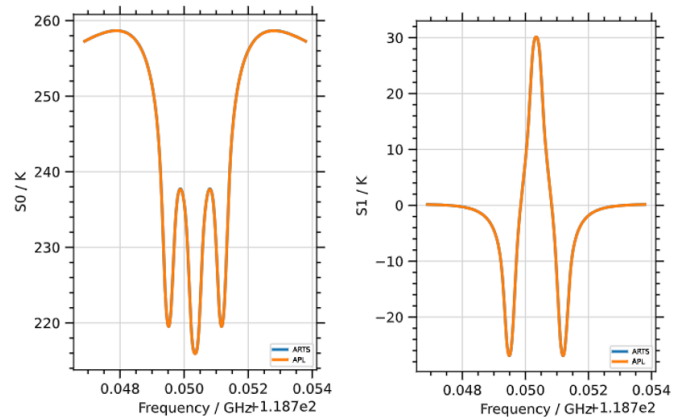


Figure 4. Simulated spectra of the first two Stokes vector elements in the presence of a magnetic field. S_0 (left) shows the total intensity with Zeeman splitting clearly visible in the line structure, while S_1 (right) demonstrates the polarization signature arising from the difference between vertical and horizontal polarization.

signatures.

2. Data & Methods

2.1. Data

In order to derive magnetic field perturbations we need to employ a wide range of datasets, from the raw satellite data to ancillary model simulations.

2.1.1. EZIE Data

EZIE collects data continuously, but not all observations correspond to Earth-viewing observations. The raw satellite data are commonly referred to as Level 0 (L0). Important data that is included in L0:

- Electronic counts (current)
- Viewing geometry (pointing direction of each radiometer)
- Satellite orientation data (pitch, roll, yaw)

We can then identify the sections where the radiometers are facing the Earth and to focus only on the radiances measured during those observations. As during some portions of the data collection process, the satellite orients itself to face away from the Earth for the purpose of calibration. An example of multiple EZIE orbits is shown in Figure 5. We want to ignore non-science data samples associated with space views (low S_0). Additionally, are gradients and spikes are evident in S_0 (e.g., around observation index 800), right before returning back to nominal (i.e., higher) S_0 values. These data filtering techniques are also useful in the future, as there may be other space weather activities that may be negatively influencing our retrieval process.

By analyzing the pitch, roll, and yaw components, we can identify sections where the satellite maintains the

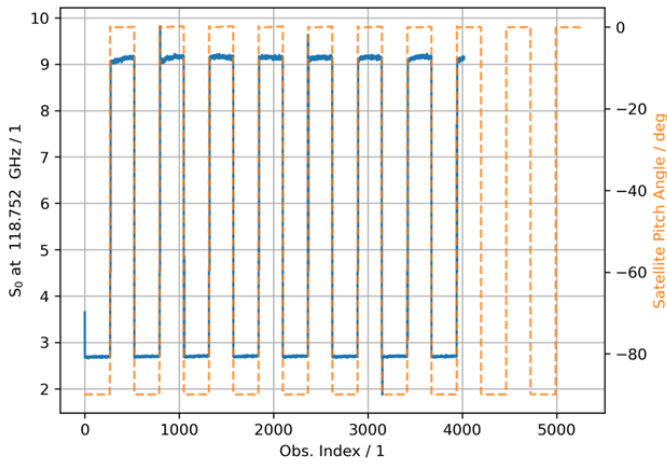


Figure 5. Time series of S_0 (blue, left axis) and satellite pitch angle (orange, right axis) across multiple EZIE orbits. The data highlights intervals when the satellite faces Earth, during which higher radiances are measured, versus periods when the satellite is oriented away from Earth for calibration. Small spikes in S_0 , such as near observation index 800, mark transitions between these orientations. Identifying these patterns is important for focusing on valid radiance measurements.

same orientation for an extended period of time without spacecraft adjustments. As shown Figure 5, there are portions where the satellite pitch and the collection of the data overlaps, which can introduce uncertainty in the sampled signals. In our Method's section we will discuss how to resolve this.

The raw satellite data (i.e., the electronic counts in the L0 data files) are used to derive calibrated Stokes vector radiances, which are called Level 1 data (L1). Stokes vectors have 4 components: the vertical polarization, the horizontal polarization, S_3 and S_4 represent the difference between two linear polarizations, as well as the rotational polarization, respectively.

Figure 6 shows a comparison of simulated (orange) and observed (blue) Stokes vector elements for a single observation point. However, we do see that the simulated results do not match the measurements. This is a result of a misaligned vertical polarization axis of the EZIE radiometers and, as a result, the observed Stokes elements will not match what our simulations predicted.

To correct for this misalignment, we developed a method to apply a gradual rotation angle, which is further discussed in the Method section of this paper.

Simulating the EZIE observations is a complicated procedure. We need to incorporate data sets from IGRF, NRLMSIS, and MERRA-2. These are ancillary model data sets that try to replicate realistic atmospheric conditions for different variables. These data sets are described in more detail below.

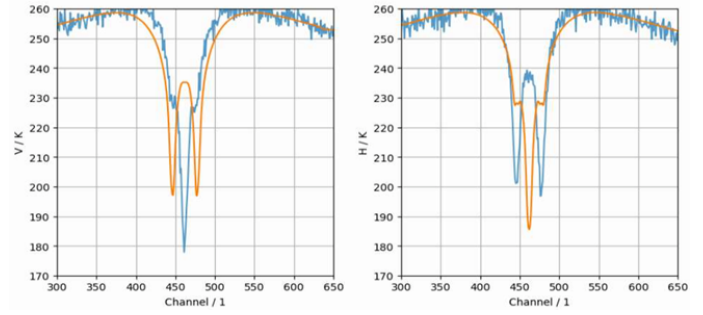


Figure 6. Comparison of vertical (V, left) and horizontal (H, right) Stokes components collected from the EZIE satellite (blue) and simulated results (orange) on 05/07/2025. While both spectra capture the same general absorption features, discrepancies between the measured and simulated results are evident. These differences arise from geometric orientation offsets in the radiometers, which cause the observed Stokes parameters to diverge from the predictions of the forward model.

2.1.2. International Geomagnetic Reference Field (IGRF)

The International Geomagnetic Reference Field is a mathematical model of the Earth's Magnetic Field. The field slowly changes over time and IGRF is updated regularly. We utilize IGRF data in our simulation pipeline. IGRF is used to visualize the magnetic field perturbations and will also serve as a baseline when we perform our retrievals. In Figure 7, we see the magnitude of the magnetic field East component and how it is distributed globally. We use these data, along with the other directions (North and Up), to replicate the real-life magnetic that we will be using in our simulations. [4]

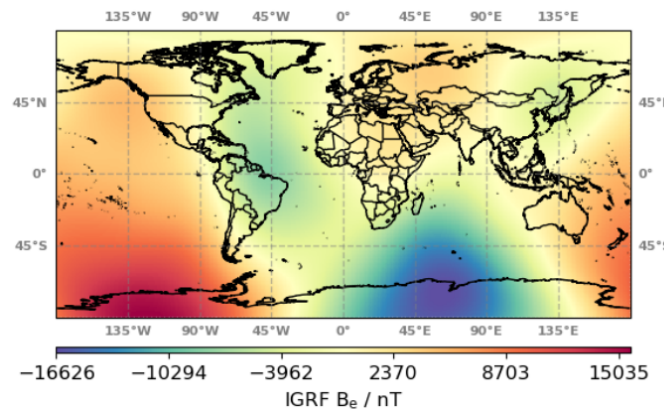


Figure 7. Global map of the eastward magnetic field component (B_e) from the International Geomagnetic Reference Field (IGRF), shown in nano-Tesla (nT). The map highlights large-scale spatial variations in the geomagnetic field, with strong eastward and westward contributions across different regions of the globe. These variations provide critical context for modeling geomagnetic perturbations and their influence on satellite observations.

2.1.3. Navel Research Laboratroy Mass Spectrometer and Incoherent Scatter Radar Model (NRLMSIS)

The Navel Research Laboratroy Mass Spectrometer and Incoherent Scatter Radar Model is a global empirical atmospheric model. NRLMSIS provides important data on the composition, temperature, and density of the Earth's atmosphere from the ground up to the exosphere. The data that is useful to us, and we need for our simulations, are concentrations of non-reactive atmospheric constituents like O_2 . In addition to those concentrations, we also use NRLMIS temperature profiles, as a function of altitude and pressure. However, we only use these temperatures for altitudes above 80 km. [5]

2.1.4. Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2)

The Modern-Era Retrospective analysis for Research and Applications Version 2 is a global atmospheric reanalysis dataset that is produced by NASA's Global Modeling and Assimilation Office. This reanalysis process combines a multitude of observations (from satellites and ground stations), together with modelling efforts. [1]

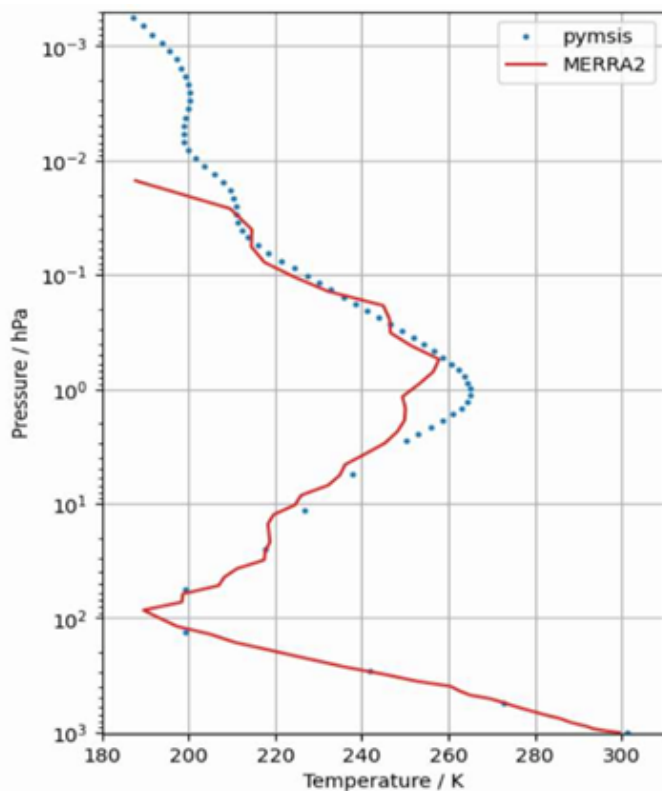


Figure 8. Atmospheric temperature profiles from NRLMSIS (blue dotted) and MERRA-2 (red). The comparison highlights structural differences across altitude ranges, particularly in the upper atmosphere where NRLMSIS extends coverage beyond the limits of MERRA-2, and in the lower and middle atmosphere, where MERRA-2 provides more detailed variability. These complementary datasets can be merged to generate realistic and continuous temperature profiles for forward modeling and retrieval applications.

Figure 8 shows an example of colocated temperature profiles from NRLMSIS and MERRA-2 for a single location and time. Differences in vertical extent and variability are evident.

2.2. Methods

This methods section is organized in a way to illustrate how we move along the data processing path to eventually retrieve magnetic fields and/or temperature profiles from using the EZIE observations. Each step is properly labeled via the respective processing level and start with L0, then L1, and ends in L2. Figure 9 shows the data processing pipeline going from the raw counts and collected EZIE data to retrieving the magnetic field vectors.

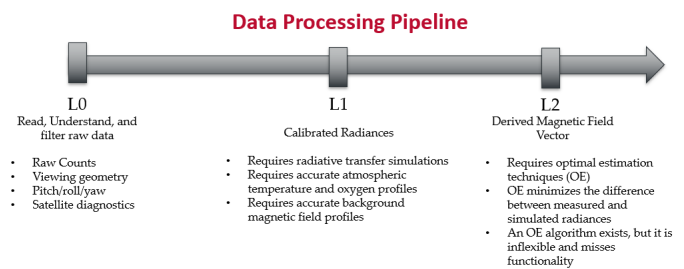


Figure 9. Processing pipeline, going from the raw EZIE observations to magnetic field vector retrievals.

2.2.1. L0

Filtering Unscientific Data

To ensure that only reliable observations make it into the final processing steps, we filter the observations by removing intervals where the satellite orientation (pitch, roll, and yaw) is changing during the process of reorientation. This process is implemented in a routine called `clean_interval.py`, which identifies different orbits, as well as “clean” intervals during each orbit where the spacecraft attitude is stable.

Overview of the Method

The core function `find_clean_intervals` takes as input the pitch, roll, yaw, latitude, and sampling time arrays. The goal is to produce a list of start and end indices corresponding to time intervals where all three orientation angles remain stable and the latitude data does not contain missing values (NaNs), i.e., geolocations not related to Earth but the space views.

Step 1: Handling Missing Latitude Data

The latitude array often contains NaNs due to calibration periods where the satellite is not looking at Earth. The helper function `split_by_nan` detects these NaNs and divides the time series into continuous, valid segments. Only these valid latitude windows are passed forward for further filtering.

Step 2: Identifying Stable Segments in Orientation Angles

Within each valid latitude window, the pitch, roll, and yaw data are examined using the `find_largest_stable_segment` function. This routine compares the orientation values to their median and applies a deviation threshold (default 0.05). The largest contiguous segment where values remain within this tolerance is selected as the stable interval.

Step 3: Combining Pitch, Roll, and Yaw

Because the satellite attitude must be simultaneously stable in all three axes, the algorithm computes the maximum of the three start indices and the minimum of the three end indices. This ensures that the retained interval corresponds to the overlap where pitch, roll, and yaw are all stable. The final result is a list of index pairs that mark the beginning and end of each “clean” interval.

Output

The output of `find_clean_intervals` is a list of (*start*, *end*) index pairs. These indices can be used to subset the latitude and orientation time series so that subsequent processing (e.g., radiative transfer simulation or retrieval alignment) uses only the most reliable geophysical data.

Significance

By combining latitude data with attitude stability checks, this procedure eliminates segments where radiances and counts collected may be unreliable due to sudden spacecraft rotations or gaps in the measurements. The cleaned dataset provides a robust foundation for later steps in the retrieval pipeline, ensuring that comparisons between simulated and observed measurements are not contaminated by unstable pointing.

2.2.2. L1

Creating the Environment/Atmosphere

In order to simulate EZIE measurements, a realistic representation of the atmosphere and the geomagnetic field must be generated for the exact date and location of each EZIE sample. This task is handled by the `run_atmos` routine in `atmos.py`, which orchestrates several modules: the IGRF magnetic fields, the NRLMSIS empirical atmospheric model, interpolation routines, and MERRA-2 reanalysis fields. Together, these components generate merged atmospheric files that can be used directly in radiative transfer simulations.

Overview of the Method

The `run_atmos` function begins with Level-0b input data to determine the observation geometry. It then builds a grid of latitude, longitude, and altitude values and executes the following steps:

1. Compute the Earth’s magnetic field using IGRF.
2. Compute the neutral atmosphere with NRLMSIS.
3. Interpolate NRLMSIS outputs to match the IGRF

grid.

4. Read MERRA-2 meteorological data for temperature, altitude, and pressure.
5. Merge all three datasets into a single atmospheric object at a common horizontal and vertical grid.
6. Save results to NetCDF and generate ARTS-ready XML files.

Step 1: Magnetic Field (IGRF) The magnetic field is calculated by `igrf_calc.py`, which wraps the `ppigrf` Python module. The code:

- Converts the L0 time stamps into a Python `datetime` object.
- Constructs 3D meshes of longitude, latitude, and altitude.
- Runs IGRF in parallel using Python’s `multiprocessing` library.
- Returns eastward (B_e), northward (B_n), and upward (B_u) magnetic field components, stored in Tesla.

This provides the geomagnetic environment necessary for our radiative simulations.

Step 2: Neutral Atmosphere (NRLMSIS) Neutral constituent densities and temperature profiles are produced by `nrlmsis_calc.py`, which uses the `pymis` Python interface to the MSIS model. The code:

- Accepts solar indices ($F_{10.7}$, $F_{10.7a}$) and geomagnetic activity (A_p).
- Runs MSIS in parallel across altitudes, returning densities for O, O₂, N₂, He, Ar, H, and NO.
- Derives temperature profiles, pressures, and volume mixing ratios (VMRs).
- Interpolates to fill missing values, especially at polar latitudes.

The output is stored in an `_NrlmsisData` class, with fields transposed into an ARTS-specific format.

Step 3: Interpolation to Common Grid Because NRLMSIS and IGRF use different native grids, `nrlmsis_interp.py` resamples the MSIS output:

- If only horizontal grids differ, 2D interpolation is applied.
- If altitude differs as well, a full 3D interpolation is applied.
- After resampling, VMRs and pressures are recomputed on the new grid.

This ensures that all model components are aligned consistently.

Step 4: Meteorology (MERRA-2)

The `merra2.py` module reads NASA’s MERRA-2 reanalysis data and:

- Extracts longitude, latitude, pressure levels, altitudes, and temperature profile from `inst3_3d_asm_Np` files.
- Combines multiple time steps, if necessary, stacking arrays along the time axis.
- Reconstructs pressure fields from MERRA-2's DELP variable using the `_pressure_from_delp` helper function.

MERRA-2 provides the meteorological context for the troposphere and stratosphere (temperature, pressure) and replaces the less reliable MSIS variables in these atmospheric regions.

Output

The merged atmospheric Python object includes:

- Densities for non-reactive constituents, temperature profiles, and pressures from NRLMSIS and MERRA-2.
- Magnetic field components from IGRF.
- A common grid structure for use with ARTS simulations.

Files are saved both as NetCDF (`.nc4`) and ARTS XML, ensuring compatibility with forward modeling workflows.

Significance

By combining IGRF, NRLMSIS, and MERRA-2 into a unified product, the pipeline provides realistic atmospheric and magnetic states tailored to the EZIE observation period. This ensures that radiative transfer simulations account for both geophysical variability and magnetic influences. Without this step, retrievals would be limited by incomplete or inconsistent background data, reducing their physical accuracy and interpretability.

Unfortunately, creating this unified product ended up being rather slow. To speed up this process and allow for a faster processing time, we utilized Python's xarrays capabilities to create the merged atmosphere files.

We managed to optimize the creation of this environment/high resolution grid through parallelization and multiprocessing, as well as lazy loading the final results which allows for much more efficient data access. Before these steps, the algorithm required a high-performance workstation. After applying the changes, we can run this part of the algorithm on a standard laptop.

Vertical Alignment

In addition to modeling the atmosphere and geomagnetic field, it is necessary to correct for instrumental effects. One of the most critical factors is the alignment of the vertical polarization axis. If the orientation is misaligned between observations and simulations, the simulated Stokes vector elements will be significantly different from the actual EZIE measurements. The `vert_pol_axis.py` routine addresses this issue by applying a series of trial

rotations to the simulated Stokes vectors and identifying the rotation angle that best aligns them with observations.

Overview of the Method

The function `vert_pol_axis` iterates over a set of candidate rotation angles (0–180° with a user-defined step size) for each of the four radiometers of an EZIE spacecraft. For each rotation, the simulated Stokes vectors are rotated and compared to the measured spectra using an adjustable cost function (chi-squared, absolute error, mean squared error, or correlation). The angle that minimizes the discrepancy is selected as the best estimate of the radiometer's polarization axis alignment.

Step 1: Rotation of Stokes Vectors

The helper function `_rotation_matrix` generates the 4×4 transformation matrix for a given angle. Each simulated Stokes vector is multiplied by this matrix to simulate the effect of rotating the instrument's polarization axis. This ensures that the physics of polarization are preserved while testing different alignments.

Step 2: Defining the Cost Function

To evaluate agreement between rotated simulations and measurements, several cost functions are supported:

- **Chi-squared:** Measures squared differences, sensitive to overall fit.
- **Absolute error:** Uses absolute differences, robust to outliers.
- **Mean error:** Averages differences across channels.
- **Correlation:** Evaluates the similarity in shape rather than magnitude.

For each observation and rotation angle, the cost is computed across Stokes vector elements V , H , S_3 , and S_1 components.

Step 3: Optimization Strategies

Because testing all angles and observations is computationally intensive, two key optimizations are applied:

- **Vectorization:** Matrix multiplications and error calculations are performed on entire arrays at once using Python's NumPy package, reducing execution time compared to explicit loops.
- **Filtering:** The analysis is restricted to latitudes within a chosen range (e.g., 30°–55°). This reduces the dataset from thousands of Stokes vectors to only the most relevant ~250, focusing computations where alignment is most critical.

Step 4: Selecting the Best Angle

For each radiometer, the algorithm identifies the angle with the lowest error (or highest correlation). Percentile-based filtering is also used to discard poor fits and take the median of the most consistent solutions. The result is a stable and reliable estimate of the polarization axis

alignment.

Output

The output of `vert_pol_axis` is an array of best-fit rotation angles for each of the four radiometers, along with detailed error statistics. If enabled, summary plots are produced to visualize the cost function values versus rotation angle, including histograms of best angles.

Significance

By applying this alignment procedure, the simulated Stokes vectors are rotated into the same reference frame as the EZIE observations. This ensures that differences between model and data reflect physical mismatches in the atmosphere or magnetic field, rather than instrument orientation errors. Furthermore, the combination of vectorization and filtering reduces the runtime from nearly four minutes to about 25 seconds, providing both accurate and efficient alignment for use in large-scale retrievals.

2.2.3. L2

At the L2 stage of the EZIE processing pipeline, the goal is to move beyond raw radiances and forward model simulations toward retrieving the underlying geophysical parameters that drive the observations (i.e., the magnetic field vectors and the temperature profiles). Since EZIE's radiances are sensitive to both atmospheric conditions and the background magnetic field, this step requires inverting the forward relationship between atmospheric state and measurements. Originally, we used the Optimal Estimation (OE) module included in the ARTS radiative transfer package, which provided initial retrieval results for further testing. The ARTS OE module has a couple of issues: the API is limited and the modules behaves largely as a black box, it provides only a limited set of diagnostic retrieval variables, and it offers limited flexibility for adapting to EZIE's unique measurement scenarios. To address these challenges, we developed a tailored Gauss–Newton retrieval algorithm that can directly handle full or partial Stokes vectors, selectively omit problematic frequencies, and provide flexibly to retrieve not only scalar or vector magnetic fields, but also atmospheric temperature profiles. This customized approach provides greater transparency, control, and diagnostic capabilities, which is essential for robust magnetic field retrievals in the context of EZIE's demanding measurement environment. Note that the custom OE code does not replace the ARTS forward model simulations or Jacobian calculation. It only performs that OE math on reshaped and custom matrices, allowing for finer control and EZIE-specific setups.

2.3. Overview of the Method

The retrieval process requires EZIE observations and their associated covariance estimates. Covariance matrices

S_x and S_y are constructed to represent uncertainties in the atmospheric state vector and the EZIE measurement variables. The a priori state and initial guess for the magnetic field is provided by IGRF, while temperature priors are derived from the merged NRLMSIS and MERRA-2 ancillary datasets.

2.4. Step 1: Forward Model

The forward model uses ARTS to simulate radiances given the current state vector guess (x_{ret}). Along with radiances, ARTS computes the Jacobians, i.e., the sensitivity of the radiances with respect to each state variable. These Jacobians are essential for updating the retrieval using Gauss–Newton iterations.

2.5. Step 2: Gauss–Newton Update

The Gauss–Newton method is applied to adjust the state vector by minimizing the difference between simulated and observed radiances. The equation and the method we are using is based primarily on the the "Inverse Method for Atmospheric Sounding Theory and Practice" book by Clive D. Rodgers [2]. Another helpful resource was the OE module and documentation of the ARTS radiative transfer package. The update step incorporates the covariance matrices to weight the contribution of measurement and prior information, ensuring a balance between fitting the observations and maintaining physical realism.

2.5.1. Formulation

The following quantities are used in the Gauss–Newton scheme:

- x : the state vector to be retrieved (e.g., magnetic field components, temperature profile).
- x_a : the prior state vector, typically provided by ancillary data such as IGRF or NRLMSIS/MERRA-2.
- y : the measurement vector, consisting of radiances observed by EZIE.
- $F(x)$: the forward model operator, which simulates radiances from a given state using ARTS.
- K : the Jacobian matrix provided by ARTS, which contains the partial derivatives of the radiances with respect to each state vector element.
- S_ϵ : the measurement error covariance matrix.
- S_a : the prior error covariance matrix.
- n : the number of measurements.

2.5.2. Iteration Step

The Gauss–Newton method allows us to compute δx , which describes the adjustments to the state vector necessary to minimize the difference between observations and simulations.

$$\delta x = x_{i+1} - x_i \quad (1)$$

After each iteration step i , the state vector is updated according to:

$$\delta x = (S_a^{-1} + K_i^T S_\epsilon^{-1} K_i)^{-1} \left[K_i^T S_\epsilon^{-1} (y - F(x_i)) - S_a^{-1} (x_i - x_a) \right]. \quad (2)$$

This balances the fit to observations (y) with the constraints provided by the prior (x_a) and their respective uncertainties.

2.5.3. Inverse Hessian

The retrieval error covariance is given by the inverse Hessian:

$$\hat{S} = (K^T S_\epsilon^{-1} K + S_a^{-1})^{-1}, \quad (3)$$

which provides an estimate of the uncertainty in the retrieved state vector.

2.5.4. Convergence Criterion

Convergence is assessed using the condition:

$$d_i^2 = (x_i - x_{i+1})^T \hat{S}^{-1} (x_i - x_{i+1}) \ll n, \quad (4)$$

where d_i^2 measures the change in state vector relative to the retrieval error covariance. If d_i^2 becomes sufficiently small, or a maximum iteration count is reached, the retrieval iteration process is terminated.

2.5.5. Diagnostics

Several diagnostics are available from the Gauss–Newton solution:

- **Averaging kernel matrix:** $A = GK$, quantifies the sensitivity of the retrieval to the true state versus the prior. This makes use of the gain matrix G .
- **Smoothing error:** $S_s = (A - I)S_a(A - I)$, the contribution of finite vertical resolution of the observation system to the total error. $S_s = (A - I)S_a(A - I)$
- **Retrieval noise:** $S_o = GS_\epsilon G^T$, error due to random noise in the measurements.
- **Degrees of freedom for signal:** $d_s = \text{trace}(A)$, where A is the averaging kernel matrix.
- **Gain matrix:** $G = (K^T S_\epsilon^{-1} K + S_a^{-1})^{-1} K^T S_\epsilon^{-1}$, relating changes in the measurement space to changes in the state vector.

This framework ensures that the retrieval balances observational constraints with physical priors, yielding stable and interpretable estimates of the magnetic field and temperature profiles from EZIE observations.

2.6. Step 3: Iteration and Convergence

This process is repeated iteratively:

1. Generate a new forward simulation with the updated state vector.
2. Recompute model radiances and Jacobians.

3. Evaluate the cost function (χ^2) and diagnostics.

Iterations continue until d_i^2 falls below a defined threshold or a maximum number of iterations is reached.

2.7. Output

The retrieval outputs the best-fit magnetic field and temperature profiles that reproduce the EZIE observed radiances within the limits of measurement and model uncertainty. Alongside the retrieved state vector, retrieval diagnostics such as error covariances and convergence statistics are provided to assess retrieval quality. These outputs form the foundation for interpreting magnetic perturbations in auroral electrojet regions.

3. Results

3.1. L0

The application of quality-flag filtering to the Level-0 (L0) data produced a substantially cleaner dataset. As illustrated in Figure 10, the raw orbit measurements initially contained both valid and invalid observation segments, with large portions at each end of the orbit flagged due to spacecraft rotation or instrument calibration. After screening, these non-scientific intervals were removed, leaving behind a continuous record of valid observations.

This cleaning process has two major impacts. First, it reduces the overall volume of data to be processed in later stages, lowering both storage and computational requirements. Second, it improves the scientific reliability of the dataset by ensuring that subsequent analyses are not contaminated by calibration periods or unstable spacecraft pointing. In practice, this means that L1 and L2 products can be derived more efficiently and with greater confidence in their physical interpretation.

3.2. L1

3.2.1. Merged Atmosphere

We improved our routines to generate merged atmospheric files that combine realistic oxygen volume mixing ratios, temperature profiles, and colocated background magnetic fields. An example merged temperature profile is shown in Figure 11, where MERRA-2 provides temperatures up to 80 km, followed by a smooth transition into NRLMSIS temperatures. Performance was also significantly improved, with the runtime of the merging routines reduced by nearly a factor of two. In addition, the merged files are now fully compatible with lazy-reading through Python's `xarray` package, which allows for selective loading of data subsets rather than reading entire fields into memory. This reduces both runtime and memory requirements, making the pipeline more scalable.

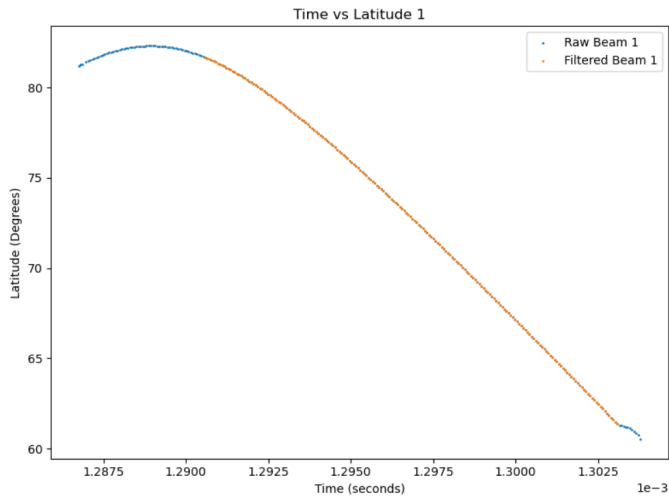


Figure 10. Time–latitude trajectory for radiometer (Beam) 1 of EZIE spacecraft "a", showing raw (blue) and filtered (orange) data. Filtering was applied using satellite pitch, roll, and yaw information to remove measurements collected during reorientation and calibration maneuvers. This ensures that only valid observations, obtained while the satellite was properly aligned, are retained for analysis.

3.2.2. Vertical Alignment

We also developed and optimized an algorithm to correctly align the vertical polarization axis of the radiometers. The method applies systematic rotations to the simulated Stokes vectors and evaluates cost functions against measured values to identify the best alignment. Figure ?? shows the corrected V and H components after finding the most appropriate rotation angle. Through vectorization and targeted filtering, the runtime of this algorithm was reduced from approximately five minutes to just 31 seconds. These improvements produce cleaner alignment results while substantially reducing computational demands.

3.2.3. Radiative Transfer Simulations

Another accomplishment are the general performance improvements in the radiative transfer simulations. By introducing parallelization and multiprocessing, the processing time to simulate full radiance spectra, across four radiometers/beams, for one observation point decreased from roughly 207 seconds to 41 seconds. These advances allow for faster forward modeling, enabling large parameter sweeps and broader diagnostic analyses within the same computational resources. Together with the improvements in atmospheric merging and polarization alignment, this results in a more efficient and reliable EZIE simulation pipeline.

3.3. L2

We successfully implemented our own OE algorithm and applied it to all 15 EZIE orbits on 04 April 2025. This con-

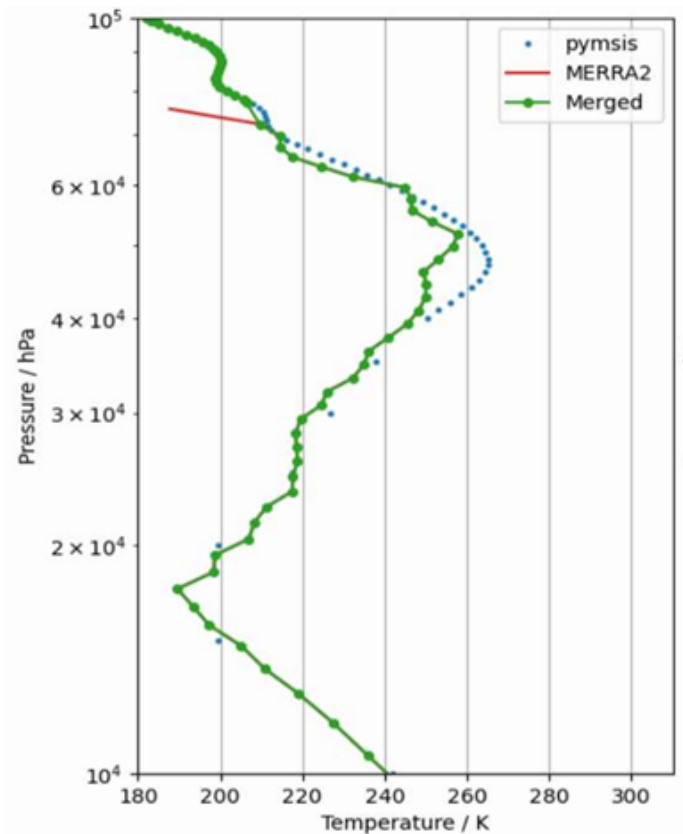


Figure 11. Example of an optimized merged atmospheric temperature profile (green) created by combining MERRA-2 (red) and NRLMSIS (blue) datasets. In this improved routine, we generate merged atmosphere files that include realistic oxygen and temperature structures along with colocated background magnetic fields. This optimized approach ensures more accurate and physically consistent profiles for use in forward modeling and retrieval simulations.

figuration included a full vector magnetic field retrieval performed simultaneously with a temperature profile retrieval. The results are shown in Figures 13-14.

3.3.1. Magnetic Field Perturbations Retrievals

During orbits 5 and 11, noticeable magnetic field perturbations were identified, where the retrieved scalar magnetic field values (orange) deviate substantially from the background IGRF values (blue). These signatures were independently confirmed by ground-based magnetometer measurements from multiple SuperMAG observing stations, providing confidence in the physical interpretation of the retrieved anomalies. In contrast, Orbits 14 and 15 showed no significant perturbations and the retrieved magnetic field results closely follow the background IGRF values. Similarly, these findings were supported by ground-based SuperMAG observations. These results demonstrate the retrieval's ability to distinguish between perturbed and unperturbed segments of an EZIE orbit.

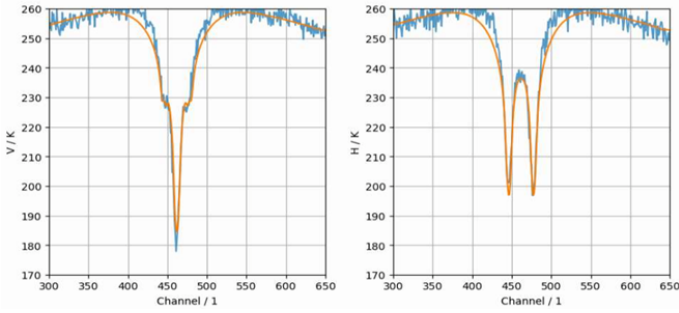


Figure 12. Comparison of simulated (orange) and observed (blue) radiances after applying polarization axis rotation. The panels show the vertical (V, left) and horizontal (H, right) polarization components following the alignment of the instrument’s vertical polarization axis. To achieve this, rotations were applied to the measured Stokes parameters in order to minimize the residuals between the modeled and observed spectra. After this correction, the V and H channels are properly aligned with the expected polarized signatures, demonstrating that the vertical polarization axis was correctly identified.

3.3.2. Consistency with Radiance Observations

We further validated the retrieval results by comparing radiative transfer simulations with the observed EZIE spectra, both before and after applying the OE algorithm. Figure 15 illustrates this comparison for one EZIE sample on 04 April 2025. All four Stokes vector elements are shown: the raw EZIE observations in blue, the forward simulations before retrieval in orange, and the simulations after the retrieval in green. The posterior spectra show excellent agreement with the measurements, demonstrating that the algorithm effectively adjusts the atmospheric and magnetic state to reproduce the observed radiances.

3.3.3. Outlook

These results mark a key milestone in the EZIE analysis pipeline, providing a robust framework for joint magnetic field and temperature profile retrievals. The EZIE team at JPL will continue to refine and expand this experimental retrieval algorithm in the coming months, with the goal of improving accuracy, robustness, and efficiency across the full mission dataset.

4. Conclusion

In this work we improved and tested a complete Python-based pipeline to support NASA’s EZIE mission, which studies auroral electrojets and small-scale magnetic field perturbations using the Zeeman-effect for radiances around the 118.75 GHz O_2 absorption line. Beginning with L0 data, we implemented robust quality screening and filtering routines to ensure that only valid scientific observations were carried forward. At L1, we constructed realistic atmospheric states by merging ancillary datasets from IGRF, NRLMSIS, and MERRA-2, while also improv-

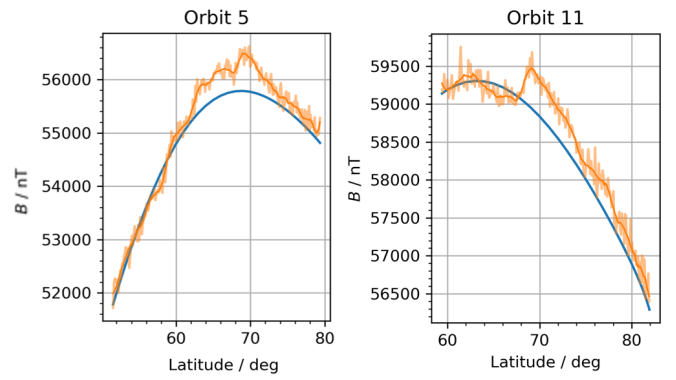


Figure 13. Retrieval results for EZIE orbits 5 and 11 on 05/04/2025, where ground-based SuperMAG measurements indicate substantial perturbations in Earth’s magnetic field.

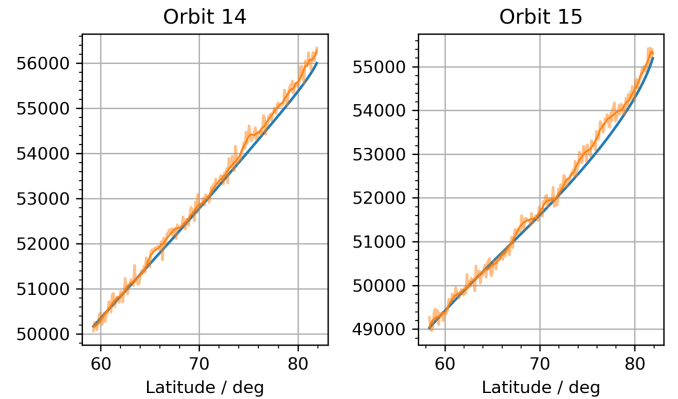


Figure 14. Retrieval results for EZIE orbits 14 and 15 on 05/04/2025, where ground-based SuperMAG measurements show no perturbations in Earth’s magnetic field.

ing radiative transfer calculations and implementing a stable vertical polarization axis alignment procedure. At L2, we introduced a Gauss–Newton inversion scheme for simultaneous retrievals of magnetic field and temperature profiles, building on the foundation of Rodgers’ inverse theory.

The results demonstrate substantial advances in both performance and scientific capability. Through careful vectorization and parallelization, the full pipeline can now run efficiently on a standard laptop, reducing run-times by factors of two to five. Simulated EZIE radiances are in close agreement with the satellite observations, and retrieval experiments successfully captured magnetic field perturbations that were independently confirmed by the ground-based SuperMAG magnetometer network. These outcomes show that the pipeline is robust and accurate enough to provide valuable scientific insights into auroral current systems.

Several key lessons emerged during this work. Parallelization and vectorization are essential to making large-scale radiative transfer and inversion simulations feasible on non-specialist hardware. The ARTS OE module and Rodgers’ *Inverse Methods* text provided crucial guidance

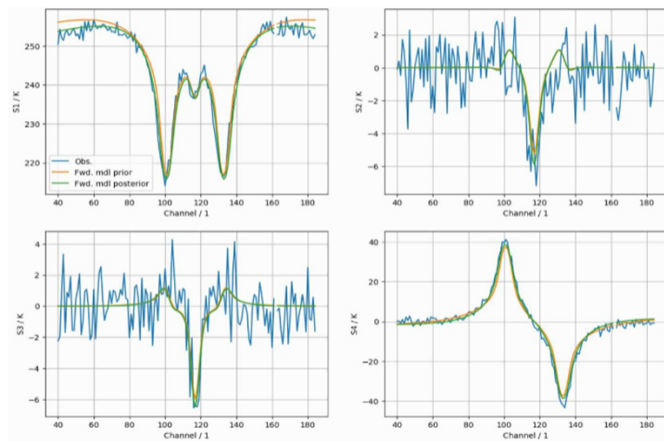


Figure 15. Example of radiance closure between observations and simulations, for one sample on 05/04/2025. Radiance spectra from all four Stokes vector elements are shown, for simulations before and after the retrieval process.

for implementing the Gauss–Newton retrieval. Finally, the performance and stability of retrievals are highly sensitive to the choice of prior information and the iteration strategy, underlining the importance of careful retrieval setups.

Overall, the improved EZIE pipeline establishes a strong foundation for future mission analysis. It enables efficient end-to-end processing from raw observations to geophysical retrievals, and its flexibility ensures that the algorithm can continue to evolve as the EZIE team at JPL expands retrieval capabilities in the coming months.

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